

Thetis-RADE User Guide -- RADE Features

Revision 1.2

This guide describes the controls Thetis-RADE adds to the standard Thetis interface to support the FreeDV RADE V1 digital voice mode.

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0. Setting up for RADE

Open Setup with the SETUP button (or via the menu), then pick the DSP tab and the RADE sub-tab. Every RADE-related setting is included here.

First enter your callsign in "Call" and your maidenhead locator gridbox in "Grid", optionally you can enter a message in "RX1Msg" and "RX2Msg".

Read the (1.) to have a complete description of all RADE controls available in Setup.

There are screenshots of currently working Setup RADE settings at the end of this document as a reference.

For first steps use only RX1 RADE.

Check RX1RADE Control

Check RX1RADE enable/disable

Check RX1Measure

Check TXMeasure

These will enable the RADE on RX1 and also the TX and relevant measurements will be on panadapter window.

For receive, in order to have proper level signal feed into RADE decoder you have to use the RX1RADE Rx level up/down control and also check the RX1Measure so to have visual indication of level or overdrive on panadapter screen.

Adjust RX1RADE Rx level so no clipping occurs and the "Level" signal's bar is a little above midway about -30 to -40 dBFS, you can config it to any value you feel proper as long as it does not clip.

For transmit, in order to have proper level signal feed into RADE encoder you have to use the RADE Mic level up/down control and also have checked the TXMeasure so to have visual indication of level or overdrive on panadapter screen.

For first transmit adjustment, click MOX (always with low power on a dummy load) and speak to microphone.

Adjust RADE Mic level so no clipping occurs and the "Level" signal's bar is below clipping with a headroom, you can configure it to any value you feel proper as long as it does not clip.

When you are ready, uncheck MOX to return to receive.

If you enable RX1RADE Loopback Test, you can have audio from speaker monitoring your microphone, it does have latency, it acts immediately without MOX, you can test/check the three RADE mic DSP conditioners.

In transmit, RNNoise/AGC/EQ of course play a significant role, so set in EQ the Vol to 0 and leave EQ unchecked to have a starting point, experiment with on/off RNNoise and AGC so to understand their action

The above are minimum but able process for a first configuration for RADE operation with Thetis-RADE.

1. Setup -> DSP -> RADE tab

Open Setup with the SETUP button (or via the menu), then pick the DSP tab and the RADE sub-tab. Every RADE-related setting is included here.

The tab is laid out top to bottom in this order: a master row of five checkboxes that govern the rest of the tab, the per-receiver RADE enable checkboxes, the loopback test checkboxes, a single mic-level trim, two per-receiver Rx-level trims, the mic conditioning chain (noise reduction, AGC and a 3-band equaliser), and the FreeDV Reporter settings.

1.1 Master row (top of tab)

There are five checkboxes on the top row. They decide whether the rest of the tab is visible at all.

****RX1 RADE Control**** is OFF by default. Tick it to bring up the RX1 RADE feature. When it is OFF, every RX1-related control on this tab is hidden and the RX1 decoder, encoder and reporter are torn down. Treat this as the master power switch for the feature.

****RX1 Measure**** is OFF by default. Tick it to display the RADE on-screen measurements (SNR, Sync, Level, Clip, last decoded callsign) on the RX1 panadapter. It is only effective while RX1 RADE is also turned on.

****TX Measure**** is OFF by default. Tick it to display the RADE on-screen TX measurements (mic Level and mic Clip) on the panadapter during transmit. It is effective when at least one of the two RADE masters is on.

****RX2 RADE Control**** is OFF by default. Tick it to bring up the RX2 RADE feature. This control is greyed out until RX1 RADE Control is on, because RX2 shares the RX1 transmit path and reporter identity.

****RX2 Measure**** is OFF by default. Tick it to display the RADE on-screen measurements on the RX2 panadapter. It is only effective while RX2 RADE is also turned on.

1.2 Per-receiver RADE enable

****RX1 RADE enable/disable**** is OFF by default. Tick it to start the RX1 RADE digital voice decoder and the transmit encoder. Received audio is decoded before reaching the speakers, and on transmit the microphone signal is encoded for over-the-air RADE transmission. The existing Thetis TX audio processing (TX EQ, Comander, CFC, Phase, Leveler) remains under your control and is applied to the mic signal before it reaches the encoder.

When this checkbox goes on, Thetis automatically sets the RX1 operating mode to DIGL below 10 MHz or DIGU at 10 MHz and above, with DIGU also being used for the 60 m band (5.0 to 5.5 MHz).

****RX2 RADE enable/disable**** is OFF by default. Tick it to start the RX2 decoder. It can run at the same time as RX1. Thetis sets the RX2 mode the same way (DIGL or DIGU based on the VFO B frequency). Transmitting on a frequency where RADE is enabled on either receiver brings up the RADE encoder.

1.3 Loopback test (diagnostic)

Both receivers offer a self-test that loops the transmit signal back into the decoder without going on the air. These are off by default and you should normally leave them off.

****RX1 RADE Loopback Test enable/disable****, when on, routes the RADE transmit signal straight into the decoder. No radio frequency is produced, because the microphone-to-radio path is silenced. This is useful for verifying that the encoder and decoder agree. Reporting is automatically turned off while loopback is on so you do not accidentally publish loopback frames.

****RX2 RADE Loopback Test enable/disable**** does the same on the RX2 path.

1.4 Mic level

****RADE Mic level (dB)**** sets the gain applied to the microphone signal just before it enters the RADE encoder. The range is -40 to +40 dB in 1 dB steps; the default is 0 dB. Use it the same way you would normally use a mic-gain trim: increase if your transmit level is too low, decrease if your peaks are too hot. While RADE is on, the standard VAC TX gain is held at 0 dB so this trim is the only mic-side gain you need to adjust.

1.5 Rx level

There is a separate decoder-input trim for each receiver. While RADE is on, the standard VAC RX gain for that receiver is held at 0 dB so the trim below is the only RX-side gain you need to adjust.

****RX1 RADE Rx level**** sets the gain applied to RX1 audio just before it enters the decoder. The range is -100 to +40 dB in 1 dB steps; the default is 0 dB. Adjust to bring the on-screen RX Level measurement (see section 3) into a comfortable range with strong signals not clipping. Remember that receiver's AGC value is also affecting this level.

****RX2 RADE Rx level**** has the same range and default as the RX1 trim and is independent of it.

1.6 Mic conditioning chain

Three optional processors sit on the microphone path in front of the RADE encoder. All three have effect only while RX1 RADE is on. They are applied in this order: RNNoise denoiser, AGC, then 3-band EQ.

****RADE Mic RNNoise enable**** is OFF by default. When on, a neural-net noise reducer cleans up the microphone signal before it reaches the encoder.

****RADE Mic AGC enable**** is OFF by default. When on, an automatic gain control follows the broadcast-style loudness standard, with a soft peak limiter at -1 dBFS. The AGC pushes your average loudness toward the target you set with the LUFS spinner.

****LUFS**** sets the AGC target loudness. The range is -30 to 0 LUFS in 1 LUFS steps; the default is -23 LUFS, which matches the freedv-gui program. More negative numbers mean a quieter target; less negative numbers mean a louder target. Most operators should leave this at the default.

****RADE Mic EQ enable**** is OFF by default. It is the master switch for the 3-band biquad equaliser described below. The post-EQ Vol control (see Vol below) is independent of this switch -- it is always active when its value is not 0 dB, even if the EQ master is OFF.

****Bass**** has a frequency setting (50 to 500 Hz in 10 Hz steps, default 100 Hz) and a gain setting (-20 to +20 dB in 0.1 dB steps, default 0 dB). Setting the gain to 0 dB turns the bass band off.

****Mid**** has a frequency setting (200 to 4000 Hz in 50 Hz steps, default 1000 Hz), a gain setting (-20 to +20 dB, default 0 dB), and a Q (bandwidth) setting (0.1 to 5.0 in 0.1 steps, default 0.7). Higher Q values make the band narrower. Setting the gain to 0 dB turns the mid band off.

****Treble**** has a frequency setting (1000 to 8000 Hz in 100 Hz steps, default 5000 Hz) and a gain setting (-20 to +20 dB, default 0 dB). Setting the gain to 0 dB turns the treble band off.

****Vol**** is the post-EQ master volume. The range is -20 to +20 dB in 1 dB steps; the default is 0 dB. It is active whenever the value is not 0, regardless of whether the EQ master is on. Use it for a broad level trim once you have shaped the spectrum with the three bands above.

1.7 Reporter (FreeDV Reporter)

The Reporter section here in Setup carries your personal identity (callsign, locator, free-form messages) and a few connection options. The Reporter window itself is described in section 5.

****RX1 RADE Reporter**** is the master switch for the FreeDV Reporter connection. Tick it to open the Reporter window and connect to the qso.freedv.org service. A single Reporter window is shared by RX1 and RX2. Untick it to close the window and disconnect.

****Ignore QSY request**** is OFF by default. When ticked, incoming QSY requests from other stations are silently recorded to the network log file instead of showing a pop-up on your screen. Enabled only while the Reporter is on.

****RX1 RADE enable reporting**** switches the Reporter between two roles. When OFF, the Reporter is in "view" mode: you see other stations but your own information is not published. When ON, the Reporter switches to "report" mode and publishes your callsign, locator, current frequency, transmit state and RADE SNR. It will only switch to report mode if both the callsign and the 6-character Maidenhead grid below are valid; otherwise it auto-unticks.

****Call**** is your station callsign as it will appear on the FreeDV Reporter map. Only letters, digits and "/" are accepted.

****Grid**** is your Maidenhead 6-character locator (for example KM18ub). The format is two letters, two digits, two letters; case is corrected as you type. Characters that do not belong at a particular position are dropped.

****RX1 Msg**** is a free-form status message broadcast alongside your RX1 identity. Common uses include rig name, antenna, or a hint about your current activity.

****RX2 Msg**** is a free-form status message broadcast alongside your RX2 identity. It is independent of the RX1 message and is visible only while RX2 RADE Control is on.

****RX2 RADE enable reporting**** is OFF by default. When ticked, it opens a second connection to qso.freedv.org that publishes the RX2 frequency and SNR independently of RX1. It is enabled only while the main RX1 Reporter is on.

2. RX1 and RX2 RADE controls on the console

For everyday operation, three compact checkboxes are placed on the console face for each receiver so you can toggle the most-used RADE options without opening Setup. These controls follow whatever you configure in Setup -> DSP -> RADE; changes propagate both ways.

2.1 RX1 cluster

Three small checkboxes appear at the bottom of the RX1 mode-specific panel.

****RADE**** turns the RX1 RADE decoder and encoder on and off. It is equivalent to the "RX1 RADE enable/disable" checkbox in Setup. When you tick it, the receiver mode is automatically set to DIGL or DIGU based on the current frequency.

****REPR**** opens and closes the FreeDV Reporter window. It is equivalent to the "RX1 RADE Reporter" checkbox in Setup. The Reporter window is shared between the two receivers, so there is no separate REPR for RX2.

****VIS**** controls whether you are visible to other Reporter users. When ON, your station is published to the Reporter network (report mode). When OFF, the Reporter still shows you

the station list but does not publish your own information (view mode). VIS is greyed out while REPR is OFF, because publishing makes no sense without an open Reporter connection.

If you turn off RX1 RADE Control in Setup, the entire RX1 cluster on the console face disappears and is disabled.

2.2 RX2 cluster

Two checkboxes appear on the RX2 panel.

****RADE**** turns the RX2 RADE decoder on and off. It is equivalent to the "RX2 RADE enable/disable" checkbox in Setup, and like RX1 it automatically sets the RX2 mode to DIGL or DIGU.

****VIS**** turns the independent RX2 Reporter connection on and off (publishing the RX2 frequency and SNR as a second station record). It is greyed unless the main RX1 REPR is on, since the RX2 Reporter connection depends on the main one.

There is no separate REPR on the RX2 panel: the single RX1 REPR covers both receivers.

3. Panadapter RADE measurements

When the panadapter Measure checkboxes (section 1.1) are turned on, Thetis-RADE draws a compact set of measurements directly on top of the spectrum in the top-right corner of the panadapter. This lets you monitor the digital voice signal without opening any other window.

3.1 When the measurements show

The overlay appears in two different forms depending on whether the radio is receiving or transmitting.

While the radio is receiving (not in MOX), each panadapter shows its own receive-side measurements -- but only if both the corresponding Measure checkbox in Setup is on AND that receiver has RADE turned on. RX1 uses the RX1 Measure checkbox; RX2 uses the RX2 Measure checkbox.

While the radio is transmitting (MOX is on), the receive overlay disappears and a smaller transmit-side overlay is shown instead on whichever panadapter the active VFO TX uses (VFO A routes to the RX1 panadapter; VFO B routes to the RX2 panadapter when RX2 is enabled, otherwise to the RX1 panadapter). The transmit overlay appears only if the TX Measure checkbox in Setup is on AND that receiver has RADE turned on.

3.2 Layout

The text is drawn in yellow, right-aligned at the top right of the panadapter.

Each value sits on top of a small coloured background bar. The cells light up in green or red to give a visual indication of the value, similar to a meter ladder.

3.3 Receive overlay (five lines)

The receive overlay shows five rows of information.

The ****SNR**** line shows the decoder signal-to-noise estimate in dB. The background bar gives a quick visual indication. When the decoder is not synchronised, the bar is empty.

The ****Sync**** line shows whether the decoder is locked onto a RADE signal. When it reads Yes, all nine background cells are green. When it reads No, the background is empty.

The ****Level**** line shows the decoder input level in dBFS, on a scale from -120 dBFS (silence) to 0 dBFS (full scale).

The ****Clip**** line warns about over-driving the decoder.

The ****Call**** line shows the callsign of the last decoded remote station.

3.4 Transmit overlay (two lines)

When the radio is transmitting, the receive overlay disappears and a smaller two-row transmit overlay is shown.

The ****Level**** line shows the microphone-input level in dBFS, with the same -120 to 0 scale and the same green-to-red cell colouring as the receive Level row. Keep the bar in the green territory; the top two red cells indicate the mic signal is approaching clipping.

The ****Clip**** line warns about over-driving the encoder. Same behaviour as the receive Clip row: green Yes/No text with an all-red background when the value is Yes, empty background when it is No.

4. Meters / Gadgets RADE measurements

The Meters / Gadgets system lets you build a custom dashboard of measurements in any layout you like. Thetis-RADE adds seven new meter types and one text gadget for RADE.

To add any of them, open the Meters / Gadgets dialog, pick a meter container, and choose the RADE entry from the meter picker.

4.1 What each meter measures

****RADE Sync**** indicates whether the decoder is locked onto a RADE signal. It is a simple on/off indicator that lights green when sync is achieved and decays softly when sync is lost (so brief drop-outs do not cause it to flash off).

****RADE SNR**** displays the decoder signal-to-noise estimate in dB on a scale from -10 dB to +40 dB.

****RADE RX Level**** displays the decoder input level in dBFS on a scale from -120 dBFS to 0 dBFS.

****RADE Clip**** is an on/off indicator that lights red the moment the decoder input peaks crossed the clip threshold. Use it as an over-level warning.

****RADE Last Callsign**** is a text gadget. It shows the callsign of the last station whose End-Of-Over (EOO) was successfully decoded. When no callsign has been decoded yet, the gadget shows dashes ("-----"). The colour of the title text and the colour of the callsign text are independently configurable from the container's colour pickers.

****RADE TX Mic Level**** is the transmit counterpart of RADE RX Level, using the same -120 to 0 dBFS scale. It shows the microphone level reaching the encoder.

****RADE TX Mic Clip**** is the transmit counterpart of RADE Clip, lighting red when the mic peak crossed the clip threshold.

4.2 Customising colours

Each bar meter exposes two colour pickers in its container settings: "Low:" for the red end of the scale, and "High:" for the green end. Pick whichever colours suit your display; the defaults (green/red) work well for most setups.

The RADE Last Callsign gadget exposes a "Title:" colour for the small heading text and an Indicator (Marker) colour for the larger callsign text.

4.3 Hide if no RADE

Each meter container in the Meters / Gadgets dialog has a "Hide if no RADE" checkbox. When ticked, the whole container disappears automatically whenever the receiver bound to that container has RADE turned off, and reappears as soon as RADE is turned back on. Use it to keep your dashboard tidy: RADE-only meters stay hidden during normal SSB operation and pop back up when you switch to RADE.

4.4 Saving and updating meter layouts

Your meter layout is remembered between sessions automatically.

If a future version of Thetis-RADE changes a meter's default appearance (for example, a different scale range or default colours), existing meters in your saved layout will keep the appearance they were created with. To refresh a meter so it picks up the new defaults, delete it from the container and add it back from the meter picker.

5. FreeDV Reporter window

The FreeDV Reporter window shows a live list of stations currently on the FreeDV Reporter network, lets you see who is calling whom and where, and lets you ask another station to move (QSY) to your current frequency.

To open the Reporter, tick "RX1 RADE Reporter" in Setup -> DSP -> RADE, or tick "REPR" on the RX1 cluster on the console face. The window can be closed and re-opened freely; closing it does not turn off the connection -- ticking the box again brings it back exactly where you left it.

The window is divided into three areas: a menu bar across the very top, a toolbar below it, the main station list in the middle, and a status bar across the bottom.

5.1 Menu bar

Two menus sit at the top of the window.

The **Show** menu lists every column available in the station list. Tick the columns you want to see; untick the columns you do not. Your choice is remembered between sessions.

The **Filter** menu controls automatic filtering of inactive stations. Pick one of the idle-cutoff options to hide stations that have not posted any update for the chosen number of minutes; pick "Disabled" to show every station regardless of idle time.

5.2 Toolbar

The toolbar holds a band-filter combo, four tracking buttons, and the Request QSY button.

The **Band** combo on the left filters the station list by ham band. Pick "All" to show every band; pick a specific band (160 m, 80 m, 40 m, etc.) to show only stations in that band. When you are tracking a receiver (see below), this combo follows the tracked receiver and snaps automatically.

Track RX1 and **Track RX2** are two mutually exclusive toggles. When **Track RX1** is on, the list filter follows the VFO A frequency: as you tune VFO A across the bands, the list updates to show stations relevant to your current frequency. **Track RX2** does the same for VFO B (it is greyed out unless RX2 is enabled on the console). Click the active button a second time to turn tracking off.

When tracking is on, the two **Frequency** and **Band** sub-mode buttons decide how the tracked frequency drives the filter. In **Band** mode (the default), the list shows every station in the same ham band as the tracked receiver. In **Frequency** mode, the list shows only stations whose reported frequency is within a small tolerance of the tracked receiver -- ideal when you are trying to spot people on the exact frequency you are listening on.

Request QSY asks the currently selected station to QSY to your current transmit frequency. Pick a station row in the list, then click Request QSY; a short network message goes out to that station and a pop-up appears on their screen (if they are running compatible software and have not disabled the pop-up) inviting them to move. The button is greyed out when no row is selected, when your transmit frequency is not yet known, when the selected row is your own station, or when the Reporter is in view mode (publishing turned off). Clicking it sends the request but does not clear your row selection, so you can quickly resend or visually confirm the target.

5.3 Station list

The middle of the window is a sortable table of stations. Click any column header to sort by that column; click again to reverse the sort order. The columns available are listed below; toggle which ones are visible from the Show menu.

- **Callsign** -- the station's published callsign.
- **Locator** -- the station's 6-character Maidenhead grid.
- **km** -- great-circle distance from your station to theirs.
- **Hdg** -- compass bearing from your station to theirs.
- **Version** -- the client software name (Thetis, freedv-gui...).
- **MHz** -- the station's reported operating frequency.
- **Mode** -- the station's reported operating mode.
- **Status** -- whether the station is currently transmitting.
- **Msg** -- the station's free-form status message.
- **Last TX** -- the time of the station's most recent transmission.
- **RX Call** -- the callsign of the last station they decoded.
- **RX Mode** -- the mode of the last signal they decoded.
- **SNR** -- the SNR they reported for that decode.
- **Updated** -- when their record was last updated.

Rows are colour-shaded to draw your eye to recent activity. A row that is currently transmitting is shaded pink/red. A row that has just decoded a signal is shaded green. A row whose free-form message just changed is shaded plum. The shading fades back to normal after a few seconds.

Click any row to highlight it in the system selection colour. The highlight stays put until you click somewhere else. Clicking on empty toolbar space, or on the Band / Track buttons, clears the selection. Clicking Request QSY uses the selection but does not clear it. Double-clicking a row tunes your VFO A to that station's reported frequency and sets the appropriate digital mode.

When the list first opens, no row is highlighted -- you must click a row to pick one. Sorting, filtering and the 1-second background refresh preserve your selection so the row you picked stays highlighted while the list updates around it.

The list automatically sticks to the top whenever the topmost row is visible, so new stations sorted toward the top stay on screen. If you scroll down to read older rows, the sticky-top behaviour disengages so you can read in peace; scroll back to the top and it re-engages.

5.4 Status bar

A three-segment status bar runs along the bottom of the window.

The left segment shows the current **Connection** state ("Connected successfully", "Communication lost", "Login attempt..." etc.). If something goes wrong it tells you here.

The middle segment shows the current **Filter** state (for example, which band is filtered, or whether idle-stations filtering is active).

The right segment shows the ****count**** of stations currently visible under the filter ("12 stations" or "1 station").

5.5 Inbound QSY pop-up

When another station requests you QSY to a specific frequency, a small pop-up appears with the requester's callsign and the target frequency in MHz. Click OK to dismiss. Thetis-RADE does NOT tune your radio automatically -- the operator decides whether to move. If you want to silence the pop-up (for example because the requests are noisy in a contest), tick "Ignore QSY request" in Setup -> DSP -> RADE; the request still gets logged but no pop-up is shown.

6. Setup RADE screenshots, controls and measurements

Setup->DSP->RADE

Setup

General Audio Display **DSP** Transmit PA Settings Appearance Keyboard Serial/Network/Midi CAT Tests

Options CW AGC/ALC AM/SAM FM Audio EER NR/ANF MNF NB/SNB VOX/DE CFC **RADE**

☒ RX1RADE Control ☒ RX1Measure ☒ TXMeasure ☒ RX2RADE Control ☐ RX2Measure

☒ RX1RADE enable/disable ☐ RX2RADE enable/disable

☐ RX1RADE Loopback Test enable/disable ☐ RX2RADE Loopback Test enable/disable

RX1RADE Rx level: -30

RADE Mic level (dB): 10

RX2RADE Rx level: -10

☒ RADE Mic RNNNoise enable

☒ RADE Mic AGC enable LUFS: -23

☒ RADE Mic EQ enable

Bass (Hz, dB): 100 0.0

Mid (Hz, dB, Q): 1500 1.8 1.0

Treble (Hz, dB): 3000 3.6

Vol (dB): 0

☒ RX1RADE Reporter ☒ Ignore QSY request

Call: SV1EIA Grid: KM18wb

RX1Msg: <https://github.com/sv1eia/Thetis-RADE> Thetis wi

☒ RX1RADE enable reporting

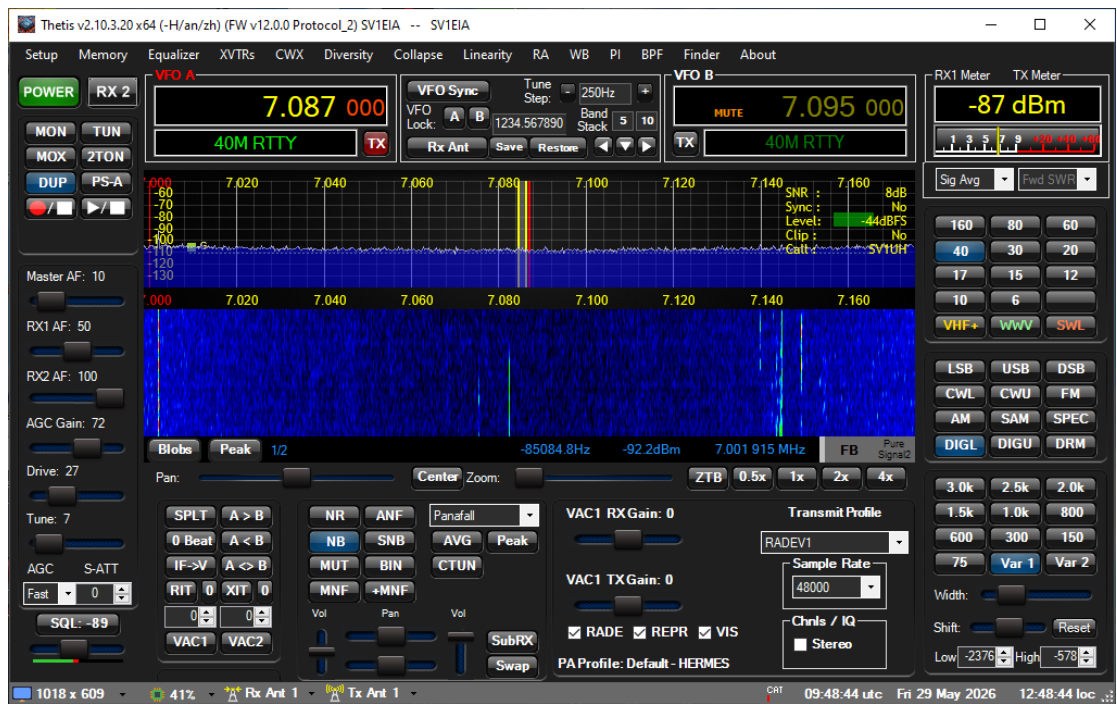
RX2Msg: Second slice

☐ RX2RADE enable reporting

OK Cancel Apply

TX Profile modified
Save profile to store

Console receive measurements on panadapter



Console transmit measurements on panadapter



Dual RADE receive, highly experimental



Setup->Appearance->Meters/Gadgets

